Baby Photoshop

Baby Photoshop is a console application (no GUI) for basic image editing, built for CS213 Assignment 1. It uses the provided `Image_Class.h` library to read, modify and write images, and everything is driven by text menus.

Team Members

Name	ID	Filters
Baher Yasser Ayad	20250114	Grayscale, Flip Image, Merge Two Images, Natural Sunlight
Seif Sabry Abd El-Salam	20250315	Black and White, Rotate Image, Detecting Image Edges, Old Den Den Mushi
Omar Wael Omar Abd Al-Latief	20250444	Invert Image, Darken & Lighten, Crop Images, Night Purple
Kirolos Nabil	20250503	Adding Frame, Resizing Image, Blur Image, Infrared

Who Did What

Baher Yasser Ayad

- Grayscale, Flip Image, Merge Two Images and Natural Sunlight filters

Seif Sabry Abd El-Salam

- Black and White, Rotate Image, Detecting Image Edges and Old Den Den Mushi filters
- File management methods

Omar Wael Omar Abd Al-Latief

- Invert Image, Darken & Lighten, Crop Images and Night Purple filters
- Project prototype and the main menu
- Input validation fixes, the `show_error_message()` and `successful_filter_message` helpers
- Fix for saving the image before loading another one
- Code formatting and the README

Kirolos Nabil

- Adding Frame, Resizing Image, Blur Image and Infrared filters
- Save and Load methods
- Validation for blur radius and resize dimensions, and color selection with custom RGB for the frame

How the Program Works

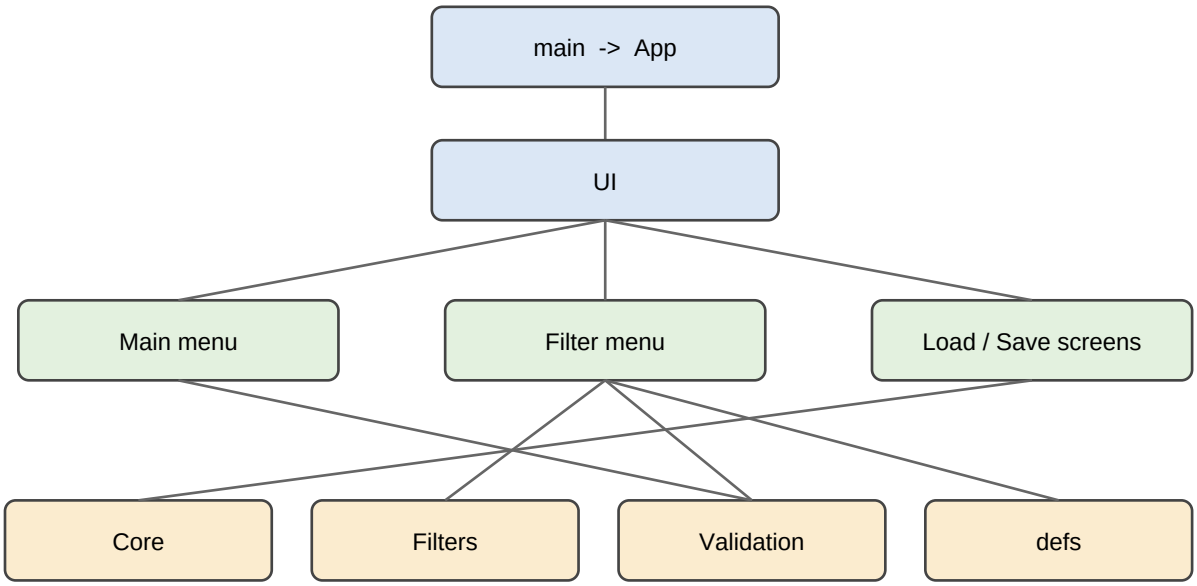
- **Main menu:** Load Image / Apply Filter / Save Image / Exit.
- **Load Image:** the user enters an image path and the image is loaded into memory. If an unsaved image is already loaded, the program asks whether to save it first.
- **Apply Filter:** the user picks one of the filters. Some filters ask for extra input (blur radius, crop points, rotation angle, frame size...). Filters can be applied one after another on the same image.
- **Save Image:** the user enters a file name, extension and path to save the edited image.

Available Filters

Grayscale, Black and White, Invert Image, Adding Frame (simple / decorative), Flip Image (horizontal / vertical), Rotate Image, Darken & Lighten, Resizing Image, Merge Two Images, Detect Image Edges, Crop Image, Blur Image, Natural Sunlight, Old Den Den Mushi, Night Purple, Infrared.

Image Skewing and Oil Painting appear in the menu but are not available yet.

System Breakdown Diagram



- **defs:** enums for the main and filter menus, and a `Point` class used to read coordinates (crop filter).
- **filters:** one section per filter, each with its own helpers and a main function that takes the image by reference and modifies it.
- **Validation:** `read_number()` and `read_choice()` keep asking until the user enters a valid number or a choice in range.
- **Core:** reading the image name, extension and path, plus `load_image()` and `save_image()`.
- **UI:** the main menu, the filter menu, and the load / save screens.
- **App / main:** starts the program by showing the main menu.

GitHub Snapshots

Commits on Sep 29, 2026

- **Some files has been formatted!!!** — OmarWael1911 committed 2 weeks ago — `233d793`
- **Add Frame, Blur, Resizing and Infrared filters has been added!!!** — kirolos500 committed 2 weeks ago — `4d12cfe`

Commits on Sep 28, 2026

- **Crop Images, Darken & Lighten, Invert Image and Night Purple filters has been added!!!** — OmarWael1911 committed 2 weeks ago — `2948828`
- **Black & White, Old Den Den Mushi and Rotate Image filters has been added!!!** — seifsabry2007-alt committed 2 weeks ago — `03c7244`
- **Main menu has been added!!!** — OmarWael1911 committed 2 weeks ago — `8bce9f2`
- **File management methods has been added** — seifsabry2007-alt committed 2 weeks ago — `701524e`

Commits on Sep 26, 2026

- **Save and Load methods has been added** — kirolos500 committed 2 weeks ago — `f0e7cd4`
- **Prototype for Baby Photoshop Project!!!** — OmarWael1911 committed 2 weeks ago — `6e1e2c9`

Commits on Oct 1, 2026

- **Fix: Crop image filter's input validation issues** — OmarWael1911 committed 2 weeks ago — `020ff40`
- **Fix: Crop image filter's input validation issues** — OmarWael1911 committed 2 weeks ago — `29ee9e4`
- **Merge branch 'main' of https://github.com/OmarWOmar19/baby_photoshop** — OmarWael1911 committed 2 weeks ago — `6dd10ff`
- **Fix: some formatting issues** — OmarWael1911 committed 2 weeks ago — `9afcf98`
- **Fix save image before loading another one issue!!!** — OmarWael1911 committed 2 weeks ago — `c00707c`
- **Change output header include "ui/output.cpp" to output source** — OmarWOmar19 authored 2 weeks ago — Verified — `766e61d`
- **Merge branch 'main' of https://github.com/OmarWOmar19/baby_photoshop** — OmarWael1911 committed 2 weeks ago — `08858c2`
- **Fix validation issues & Enhance some resuablity "successful_filter_message"** — OmarWael1911 committed 2 weeks ago — `5927424`
- **Detect Image Edges filter has been added!!!** — seifsabry2007-alt committed 2 weeks ago — `beb938b`

Commits on Sep 30, 2026

- **Grayscale, Flip image, Natural Sunlight and Merge Images Filters has been added!!!** — BaherYasser07 committed 2 weeks ago — `d91aa7c`

Commits on Oct 11, 2026

- **Fix: change section number.** — OmarWOmar19 authored 2 days ago — Verified — `86e0b88`
- **README.txt file has been added!!!** — OmarWael1911 committed 2 days ago — `ade1827`

Commits on Oct 10, 2026

- **Add color selection and custom RGB to frame filter** — kirolos500 committed 2 days ago — `28d8306`
- **Validate resize dimensions to prevent invalid image size** — kirolos500 committed 2 days ago — `d271a94`
- **Validate blur radius to prevent division by zero** — kirolos500 committed 2 days ago — `d515052`

Commits on Oct 5, 2026

- **Fix: spling for 'invalied' to be 'invalid' & Enhance: resuablity by create 'show_error_message()' function.** — OmarWael1911 committed last week — `21037dd`

Best & Hardest Part of the Assignment

- Best part of this project was building the structure of it form the beginning and it is also the hardest part.